

Monica Joya
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Assignment 01
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Question 1 - *What is the definition of a telecommunications system, and what is its primary scope in modern communication?*

A telecommunications system is a communication system that exchanges information across distances using electronic means (Slide 7), and it is made up of three core components: a transmitter, a channel, and a receiver (Slide 14). Its primary scope in modern communication covers both wired and wireless networks, the internet, satellite systems, broadcasting, and mobile communications (Slide 9). A mobile phone call demonstrates this, as the caller's voice is carried as a signal across the network infrastructure to reach the listener on the other end (Slide 25).

Question 2 - *Explain the basic components of a communication system and describe the role of each: transmitter, channel, and receiver.*

A communication system is built from three basic components that operate in sequence: a transmitter, a channel, and a receiver (Slide 14). The transmitter is the device or system that sends the signal (Slide 15). It takes the original message and encodes it into that signal for transmission, the way the microphone and circuitry in a mobile phone convert a voice into an electrical signal (Slides 15, 26). The channel is the transmission medium that carries the signal from one end to the other, and it can be wired, such as fiber or copper, or wireless, such as radio or satellite (Slide 18). The receiver is the device or system at the far end that receives the signal and decodes it back into the original message, such as the speaker and circuitry in the phone on the other side (Slides 21, 26). Together these three form the complete communication system, flowing in order from transmitter to channel to receiver, with noise affecting the signal along the way (Slide 23). A mobile phone call shows the full chain in action, as the caller's voice is encoded and sent across the network infrastructure and then decoded back into sound for the listener on the other end (Slide 25).

Question 3 - *Differentiate between analog and digital communication systems. What are the key performance metrics used to evaluate them (e.g., bandwidth, noise, latency)?*

An analog communication system uses continuous signals, a smooth wave that varies over time such as FM radio, while a digital communication system uses discrete signals, the 0s and 1s seen in computer data (Slides 35, 36, 38). The deck compares the two on three points, noise sensitivity, efficiency, and clarity (Slide 41), and gives the practical example of analog TV versus digital streaming (Slide 42).

Both are evaluated using four key performance metrics (Slide 40). Signal-to-noise ratio (SNR) compares the strength of the wanted signal to the background noise, bandwidth is how much data the link can carry, latency is the delay before data arrives, and throughput is how much data actually gets through over time.

Question 4 - *Describe the main types of communication (voice, data, video, multimedia) and explain how network topology and system architecture influence their transmission.*

There are four main types of communication. Voice is the transmission of spoken words over telecommunication systems, as in telephony and VoIP (Slide 27). Data is the exchange of digital data between devices, such as internet and email (Slide 28). Video is the transmission of moving images, such as video calls and streaming (Slide 29). Multimedia integrates voice, data, and video, as in Zoom and WhatsApp (Slide 30).

System architecture sets how the devices are connected and topology sets the layout of those connections (Slides 43, 44). That layout determines a network's reliability, manageability, and cost, since a bus or ring fails if a single point breaks, a star is easy to manage but depends on its central hub, and a mesh is the most reliable but the most expensive (Slides 45 to 48). Because voice, data, video, and multimedia all depend on bandwidth and low latency (Slide 33), the layout and architecture chosen decide how dependably and at what scale that traffic moves, so real networks blend topologies, such as a star combined with a mesh (Slides 49, 50).

Question 5 - *How do different network topologies (such as star, mesh, ring, and bus) impact the performance, reliability, and scalability of a communication system?*

Network topologies trade off most clearly in reliability, and the deck frames each one by how it fails. A bus runs on a single shared backbone, simple to build but fragile, since one backbone failure brings the whole network down (Slide 45). A star routes every device through a central hub, easy to manage but dependent on that one point, so if the hub goes down everything attached goes with it (Slide 46). A ring passes the signal device to device around a loop, which keeps its behavior predictable, but a single break disrupts the whole loop (Slide 47). A mesh connects every node to every other, the most reliable through its redundant paths but also the most expensive, and so the hardest to scale on cost (Slide 48). For scalability the deck does not single out any one of these, instead pointing to the hybrid approach, which blends layouts to stay scalable and flexible, the reason enterprise networks often combine a star with a mesh (Slides 49, 50).

Using an academic textbook on networking, I learned the bridge between a topology's layout and its performance. Performance comes down to throughput and delay, and congestion raises delay (Forouzan, 2007, pp. 7-8). The layout sets this through the type of connection: mesh, star, and ring use dedicated point-to-point links that reserve each link's full capacity, while a bus is multipoint, sharing one backbone whose capacity is split among all its devices (Forouzan, 2007).

The same textbook also completes the scalability picture. The slides went as far as the mesh's cost and named the hybrid as the scalable choice, so the textbook fills in the other three. A star grows easily, since each device needs only one link to the hub, so adding, moving, or removing a device means changing a single connection (Forouzan, 2007, pp. 10-11). A ring is also fairly easy to grow, because adding or deleting a device changes only two connections, limited mainly by how long the ring can be and how many devices it holds (Forouzan, 2007, pp. 12-13). A bus grows poorly by contrast, because there is a cap on how many taps it can carry and how far apart they sit, and since it is built to run efficiently from the start, adding devices later is difficult (Forouzan, 2007, pp. 11-12).

Reference

Forouzan, B. A. (2007). Data communications and networking (4th ed.). McGraw-Hill.